Siba Smarak Panigrahi

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Education

Ph.D. in Computer Science — **2024 – Ongoing**
École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland — *GPA: 5.67/6.0*

M.Sc. (Thesis) in Computer Science — **2022 – 2024**
McGill University and Mila, Montréal, Canada — *GPA: 4.0/4.0*
Thesis: Equivariance in the Era of Large Pretrained Models (Supervisor: Siamak Ravanbakhsh)

B.Tech. in Computer Science and Engineering — **2018 – 2022**
Indian Institute of Technology, Kharagpur, India — *GPA: 9.73/10*
Department Rank **2** in graduating batch of CSE students (Supervisor: Abir Das, Rameswar Panda).

All India Senior School Certificate Examination (AISSCE) — **2018**
Kendriya Vidyalaya Sangathan (KVS), India — *98.6%*
Secured AIR 3, Rank 1 in Bhubaneswar Region. Among the top 0.1% of the 1.16 million candidates.

Research Interests

Improving LLM agents for scientific tabular datasets, RL for LLMs, AI for Science

Conference Publications

[10] **Unsupervised Process Reward Models** (Paper)
A. Gadetsky*, M. Kodryan*, **S. S. Panigrahi**, H. Guo, M. Brbić
*Under review at International Conference on Learning Representations (**ICLR**) 2027.*

[9] **HeurekaBench: A Benchmarking Framework for AI Co-scientist** (Paper | Code | Website)
S. S. Panigrahi*, J. Videnović*, M. Brbić
*International Conference on Learning Representations (**ICLR**) 2026.*

[8] **BigDocs: An Open and Permissively-Licensed Dataset for Training Multimodal Models on Document and Code Tasks** (Paper | Dataset | Website)
J. A. Rodriguez*, X. Jian*, **S. S. Panigrahi***, et al.
*International Conference on Learning Representations (**ICLR**) 2025; **NeurIPS** 2024 - RBFM workshop.*

[7] **SymmCD: Symmetry-Preserving Crystal Generation with Diffusion Models** (Paper | Code)
D. Levy*, **S. S. Panigrahi***, S-O. Kaba, Q. Zhu, K. Lee, M. Galkin, S. Miret, S. Ravanbakhsh
*International Conference on Learning Representations (**ICLR**) 2025; **NeurIPS** 2024 - AI4Mat (spotlight).*

[6] **Improved Canonicalization for Equivariant Adaptation of Large Pretrained Models** (Paper | Code)
S. S. Panigrahi, A. K. Mondal
***CVPR** 2024 - Workshop on Equivariant Vision (**EquiVision**).*

[5] **Efficient Dynamics Modeling in Interactive Environments with Koopman Theory** (Paper)
A. K. Mondal, **S. S. Panigrahi**, S. Rajeswar, K. Siddiqi, S. Ravanbakhsh
*International Conference on Learning Representations (**ICLR**) 2024; **EWRL** 2023.*

[4] **Equivariant Adaptation of Large Pretrained Models** (Paper | Blogpost @ ICML'24 | Code)
A. K. Mondal*, **S. S. Panigrahi***, S-O. Kaba, S. Rajeswar, S. Ravanbakhsh
*Neural Information Processing Systems (**NeurIPS**) 2023, EquiVision (**CVPR**) 2024 (spotlight).*

[3] **Identifying bot activity using topology-aware techniques** (Patent)
G Choudhary, SI Rahaman, **S. S. Panigrahi**, P Bhutani, M Kilaru, K Singh, A Singhania
US Patent 2023.

[2] **[Re]: Value Alignment Verification** (Paper | Code)
S. S. Panigrahi*, S. Patnaik*
*ML Reproducibility Challenge (**MLRC**) 2021; **NeurIPS 2022 Spotlight** and Journal Showcase Track.*

[1] **Leveraging Pre-trained Language Models for Key Point Matching** (Paper | Code)
M. N. Kapadnis*, S. Patnaik*, **S. S. Panigrahi***, V. Madhavan*, A. Nandy
***EMNLP** Workshop - Workshop on Argument Mining, 2021.*

Journals

[2] **A Gene-Program Architecture of Mouse T cells** (Paper | Code)
Z. Zhang, T. Wang, **S. S. Panigrahi**, et al.
Manuscript under consideration at Nature Immunology 2026.

[1] **immgenT: A Comprehensive Reference of Convergent T-cell States in the Mouse** (Paper | Code)
I. Magill, O. Casey, D. Mallah, **S. S. Panigrahi**, et al.
*Accepted in Principle (AIP) at **Nature Immunology** 2026.*

Industry Research Experience

Document Understanding with LLMs — **ServiceNow Research**
Visiting Researcher | Multimodal Foundation Models team — *Mar 2024 - Jul 2024*
Improved **document understanding** capabilities of pretrained **LLMs** on a permissively licensed open dataset [P8].

Diffusion Models for Materials Generation — **Mila/Intel Labs**
Supervisor(s): Siamak Ravanbakhsh, Santiago Miret, Michael Galkin — *Sep 2023 - Aug 2024*
Trained **conditional diffusion models** for **materials generation** and **crystal structure prediction** task. Proposed meaningful representations for space groups and atom site symmetries for conditioning and **interpretability** [P7].

Contextual Bias in Visual Recognition Models — **IIT Kharagpur/MIT-IBM Watson AI Lab**
Supervisor(s): Abir Das, Dr. Rameswar Panda — *Apr 2021 - Dec 2021*
Evaluated **mAP** and used **GradCAM** with state-of-the-art computer vision models to quantitatively and qualitatively determine the contextual bias in images containing exclusive and co-occurring biased pairs. Proposed shallow models to capture contextual bias and **knowledge distillation** approaches for automated bias mitigation to improve.

Improving Digital Marketing with Topological Data Analysis — **Adobe Research**
Research Intern | Supervisor(s): Iftikhar Ahamath Burhanuddin — *May 2021 - Jul 2021*
Implemented **Topological Regularization** in **LSTM** Encoder-Decoder architecture to leverage topological information from customer navigation patterns and provide insights on customer sessions. Resulted in a published US patent [P3].

Academic Achievements & Honors

- **Eastern European Machine Learning** (EEML) Summer School (with travel scholarship) (2026)
- **AI for Science European Summer School** (with travel scholarship) (2026)
- **Swiss AI Initiative PhD Fellowship** for PhD students in Switzerland (2025)
- **Amii Upper Bound** bursary to attend Upper Bound AI conference (2025)
- **EDIC Fellowship** for starting Ph.D. at EPFL (2024)
- **Mitacs Accelerate Fellowship** at ServiceNow Research (2024)
- **Graduate Research Enhancement and Travel** (GREAT) Award, McGill University (2024)
- **Google's CS Research Mentorship Program** (CSRMP) Class of 2023b (2023)
- **CIFAR Deep Learning + Reinforcement Learning** (DLRL) **Summer School** (2023)
- **The Cornell, Maryland, Max Planck Pre-doctoral Research School** (CMMRS) (2023)
- **Jamsetji Nusserwanji Tata** (JN Tata) **Scholar** (endowment for higher studies) (2022)
- **Eastern European Machine Learning** (EEML) **Summer School** (selected for poster presentation) (2022)
- **Research Week with Google (Computer Vision track)** (1 of 150 selected students) (2022)
- **Prof. J.C. Ghosh Memorial Endowment Prize** (highest CGPA after semester VI) (2021)
- **Indo-US Science and Technology Forum (IUSSTF)** - **Viterbi** Award (1 of 15 awardees) (2021)
- **DAAD-WISE scholarship**, **University of Freiburg** (declined) (2021)
- **Technology Alumni Association (Delhi Chapter) Award** (highest CGPA after semester II) (2019)
- **Jagadis Bose National Science Talent Search Examination** (Rank 2 of 173 awardees) (2018)
- **KVS Junior Mathematical Olympiad** (Rank 6 in India; Rank 1 in Bhubaneswar region) (2016)
- **Exchange Student, Sakura Exchange Program in Science** (1 of 90 selected students) (2016)

Relevant Coursework

Mathematics - Linear Algebra, Calculus, Probability, and Statistics
Computer Science - Training Large Language Models, Geometry and Generative Models (GGM), Probabilistic Graphical Models, Network Science, Causal Inference and ML, Deep Learning, Machine Learning, Reinforcement Learning, Natural Language Processing (NLP), Information Retrieval, Data Visualization, Computer Networks*, Operating Systems*, Algorithms - I* & II, Cryptography & Network Security, Theory of Computation, Compilers*, Software Engineering* (* includes lab component)

Skills

- **Programming Languages** - Python, C, Java, LaTeX, Verilog, MIPS
- **Libraries** - PyTorch, PyTorch Lightning, PyTorch Geometric, NeMo Gym, NeMo RL, Huggingface, Hydra
- **Web Development** - Django, HTML, CSS, Bootstrap, PostgreSQL

Professional Services

- **Organizing**:
  - ICBINB: Failure Modes of AI in Biology workshop at **NeurIPS 2026**
  - ML Reproducibility Challenge (MLRC) 2023
  - CampusPulse (sponsored **CA$ 6000** GCP credits)
  - Molecular ML Conference (MoML) 2024, MoML 2023
- **Posters/Talks**:
  - Eastern European Machine Learning (EEML) Summer School, Cetinje (Poster, 2026)
  - Human Cell Atlas (HCA) General Meeting, Boston (Poster, 2026)
  - International Conference on Learning Representations (ICLR), Rio de Janeiro (Poster, 2026)
  - Computer and Communication Sciences (EDIC, EPFL) Open House, Lausanne (Poster, 2026)
  - International Conference on Learning Representations (ICLR), Singapore (Posters, 2025)
  - EquiVision workshop, Computer Vision and Pattern Recognition (CVPR), Seattle (Talk, 2024)
  - Google Research, Bengaluru (Talk, 2024)
  - International Conference on Learning Representations (ICLR), Vienna (Poster, 2024)
  - Conference on Neural Information Processing Systems (NeurIPS), New Orleans (Poster, 2023)
  - European Workshop on Reinforcement Learning (EWRL) 2023, VU Brussel (Poster, 2023)
  - NeurIPS 2022 Journal Showcase Track (and Spotlight lightning talk) (Poster & Talk, 2022)
  - 50th Anniversary of School of Computer Science, McGill University (Poster, 2022)
  - KDAG Winter Workshop, lecture on Support Vector Machines (Talk, 2020)
- **EDIC, EPFL Admission Committee**: Responsible for identifying strong PhD applicants (for Fall 2026)
- **Teaching Assistant (TA, EPFL)**: CS401 (Head TA, Autumn 2025, 600+ students), CS250 (Spring 2025)
- **Volunteering**: DNetCV 2022 (CVPR-Workshop), EMNLP 2021
- **Reviewer**:
  - TMLR
  - ICML 2026 (Gold Reviewer, Free registration)
  - NeurIPS 2026, 2025, 2024
  - ICLR 2026, 2025
  - AI4Mat (ICLR 2025, NeurIPS 2024 Workshop)
  - GRaM (ICLR 2026, ICML 2024 Workshop)
  - NeurReps (NeurIPS-Workshop 2024, 2023)
  - MoML (2024, 2023)
  - MLRC 2022 (Outstanding Reviewer)
  - DNetCV (CVPR-Workshop 2022)
- **Other**:
  - Member, Mental Health Committee, Mila
  - Head, Kharagpur Data Analytics Group (KDAG), IIT Kharagpur (organized reading-sessions)
  - Head, Institute Wellness Group, IIT Kharagpur
  - Head Boy, Student Council, KV No.2 CRPF Campus, Bhubaneswar